

T&M Tools for Field Engineers

Being an electronics/electrical field engineer is a challenging job, and for a person engaged in this job T&M tools are just like weapons to a warrior in the war field. In many cases in India, people rely on 'jugad' tools for a lot of test and measurement activities, but those are neither safe, nor accurate. So presented here are all the must-have tools for an electronics field engineer to perform his jobs safely and accurately



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Field engineers can work in a variety of jobs. They are specialists employed at companies that offer services to clients, and usually work in the field, i.e., locations owned by the client and not by the company at which they are employed. The locations can range from various production facilities and plants to oil fields. In the field, they can act as service representatives, oversee operations, install equipment and maintain and repair the existing equipment or supervise all engineering operations.

A lot of times in India, we see field engineers with strange tools such as

a bulb holder connected to two wires for testing the continuity. These tools may do the job but are not safe and accurate. So here we present some basic tools for field engineers and discuss why and how they are used.

Clamp meter

Flow of electrical charge (current) is used to power everything from appliances at home to huge machinery at various industries. In cases where electricity-powered equipment need to be repaired, serviced, maintained or installed, clamp meters can be indispensable tools. A clamp meter is a device that measures various properties of an electrical current. It is a preferable

device in the field due to its numerous advantages, including safety, efficiency, versatility and convenience. When clamp meters are used, you do not need to cut any wires, lowering the chances of fatal shocks. Clamp meters can give accurate readings, allowing engineers to quickly diagnose and resolve most problems. Fig. 1 shows a clamp meter.

A clamp meter measures current, voltage, crest factor and continuity. AC current is the primary measurement that a clamp meter should be able to perform. These measurements are taken from different branches of a distributed system to debug the issue.

If an appliance is not working, the first thing to check is the input supply. Most clamp meters are capable of reading AC and DC voltages. These meters can also read crest factor which refers to the measurement of a waveform. The crest factor value is determined by the waveform's peak value versus its RMS value and is important in determining whether a circuit is overloaded. Continuity can also be checked using a clamp meter and the same is indicated with a beeps. Table 1 shows various clamp meters available from different manufacturers.



Fig. 1: A clamp meter